Alessandro Salo

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Utrecht, Netherlands

PROJECTS

GPU Wavefront Ray-tracer with Vulkan *– C++, Vulkan, GLSL/SPIR-V, OpenCL, CMake*

- Built a Vulkan backend from scratch supporting a custom made GPU ray-tracier and GPU Linear Bounding Volume Hierarchy (using Morton codes, radix sort, hierarchy construction) in GLSL compute shaders.
- Bug fixed ray-tracing wavefront pipeline.
- Achieved real-time ray-tracing due to a **352×** rendering speed-up (5067ms to 14.4ms) and a **~9×** faster BVH construction.

Acceleration Structure Comparisons of BVH, BVH-SAH and Grid *–C++*

- Implemented and benchmarked three CPU ray-tracing acceleration structures (median-split BVH, SAH BVH, uniform grid with 3DDDA traversal) across dense, sparse, and uniformly-distributed scenes.
- Implemented and obtained metrics for per-ray traversal and intersection statistics to quantify the space- vs. object-partitioning tradeoff.

SIMD and GPU Cloth Simulation Optimization *–C++, SSE/SIMD, OpenCL*

- Parallelized a Verlet-integration cloth simulation using SSE intrinsics and OpenCL compute kernels; requiring Structure of Arrays data layout and race-condition free solutions.
- Achieved **5×** (SIMD) and **19.6×** (GPU) speed-ups over the sequential baseline while preserving simulation fidelity.

Rendering Resonance (Web Rasterizer) *–JavaScript, WebGL, GLSL*

- Built a billboard-based particle system (fire, smoke, fireflies) and normal mapping for a rasterizer compatible with bloom, fog, and toon shading.

Additional Projects

- **ML and Deep Learning:** CNN classifiers, a from-scratch YOLO detector, and fine-tuned BERTweet for sentiment classification (PyTorch, HuggingFace Transformers).
- **Monte Carlo Simulation Framework:** interactive C++ statistics framework with a custom AST-based math-expression parser.
- **3D Voxel Reconstruction:** multi-camera calibration, background subtraction, and voxel carving pipeline (Python, OpenCV).

EDUCATION

MSc Game and Media Technology - *Utrecht University* 2025 - present

- Coursework in graphics, computer animation, GPGPU, and optimization; current weighted average 8.43/10.
- Advanced Graphics 9.1, Computer Animation 8.8, Computer Vision 8.1; Optimization and Vectorization (10), and more.

MSc Computational Science and Engineering - *EPFL* 2024 – 2025

- Scientific computing using C++, machine learning, and simulation of physical systems.
- Transferred to Utrecht University.

BSc Physics and Astronomy - *Utrecht University* 2020 – 2024

- Cum laude - weighted average 8.64/10.

BSc Liberal Arts and Sciences (Honours), Major in Mathematics and Physics - *University College Utrecht* 2020 – 2024

- Magna cum laude - GPA 3.87/4.0 (250 ECTS).

RESEARCH AND EXPERIENCE

Utrecht University - Physics Research 2024

Research Project (Physics BSc Thesis) that led to a Peer-Reviewed Publication

- Ran **direct-coexistence simulations** to determine fluid–crystal phase boundaries of a binary hard-sphere mixtures.
- Built and validated the simulation and analysis code in C.
- Thesis graded 8.9/10; co-authored the resulting peer-reviewed paper in the **Journal of Chemical Physics** (doi:10.1063/5.0321591).

Utrecht University Aug 2022 – Jun 2024

Science Student Assistant (Teaching Assistant) - Utrecht, Netherlands

- Sole student tutor and TA for all University College Utrecht mathematics and physics courses across two years.
- Graded assignments and gave structured, individual feedback to undergraduate students.

CERTIFICATIONS, SKILLS AND INTERESTS

CERTIFICATIONS AND PUBLICATIONS

AI and Machine Learning: Advanced Neural Networks and Deep Learning (Grade A) *Oxford - Lady Margaret Hall* 2023

Peer-reviewed publication (co-author), direct-coexistence simulations *J. Chem. Phys. 164(12)* 2026

SKILLS

Languages: C/C++, Python, C#, Bash, GLSL, JavaScript

Graphics and Rendering: ray tracing, acceleration structures, rasterization, real-time physically-based rendering, OpenGL, WebGL

HPC and GPU: Vulkan (compute + graphics), OpenCL, SPIR-V, GLSL compute shaders, parallel algorithms, SIMD (SSE / AVX)

Scientific Computing: numerical methods, Monte Carlo simulation, statistical mechanics, NumPy

ML and Computer Vision: PyTorch, Transformers, CNNs, deep learning, OpenCV

Tools: CMake, Git, Linux, GPU profilers, GoogleTest, Doxygen, ImGui

INTERESTS AND HOBBIES

Bouldering, board games, hiking, photography, music, cooking, philosophy, literature, game development.